

****Production-Grade Automated Weekly Newsletter****

Nairobi Nexus Capstone

MERCY KIHUGU & AMOS MOKUA | 07/06/2026

The Original Problem (Week 8)

- Kamau's manual newsletter process was slow and inconsistent.
- The Week 8 automation worked but had critical weaknesses:
 - No fallback when AI failed → entire workflow died.
 - No validation of AI outputs → malformed JSON broke downstream.
 - No logging of what succeeded/failed → couldn't answer "What happened last Friday?"
 - Memory crashes on large HTML pages.

Our Goal – Production-Grade System

Make the workflow ****robust, auditable, and self-healing****:

- Structured AI output & JSON reliability
- Fault tolerance (degrade gracefully, not collapse)
- Model fallbacks (orange error output → backup AI)
- Operational logging & governance (answer any question)

Overview of Upgrades

- **Truncation** – reduce HTML to 5000 chars → no out-of-memory errors.
- **Defensive validation** – Code nodes after every AI agent; invalid items marked `valid: false`.
- **Backup Chief Editor** – triggered by orange error output; generates simpler intro.
- **QA Agent** – validates length, slang, Markdown, relevance; auto-corrects; adds a **parting thought**.
- **Governance sheet** – logs everything: sources succeeded/failed, model used, fallback triggered, QA results.
- **Audit workflow** – answers “What happened last Friday?” (or any date) in plain English.
- **Global error workflow** – sends AI-generated diagnosis when

Demo – Normal Run

1. Trigger the workflow via webhook.
2. Email arrives with witty newsletter intro and parting thought.
3. Google Sheets `Drafts` tab shows the newsletter.
4. `Governance` tab shows sources succeeded, model used, QA passed, etc.

****Result:**** Fully automated, reliable weekly output.

n8n + 🔍 📄

Overview

Personal

Chat Preview

Projects ▾

My project

Admin Panel

Templates

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Personal / Automated Weekly Newsletter Generator – Production Grade (with QA) an... + Add tag

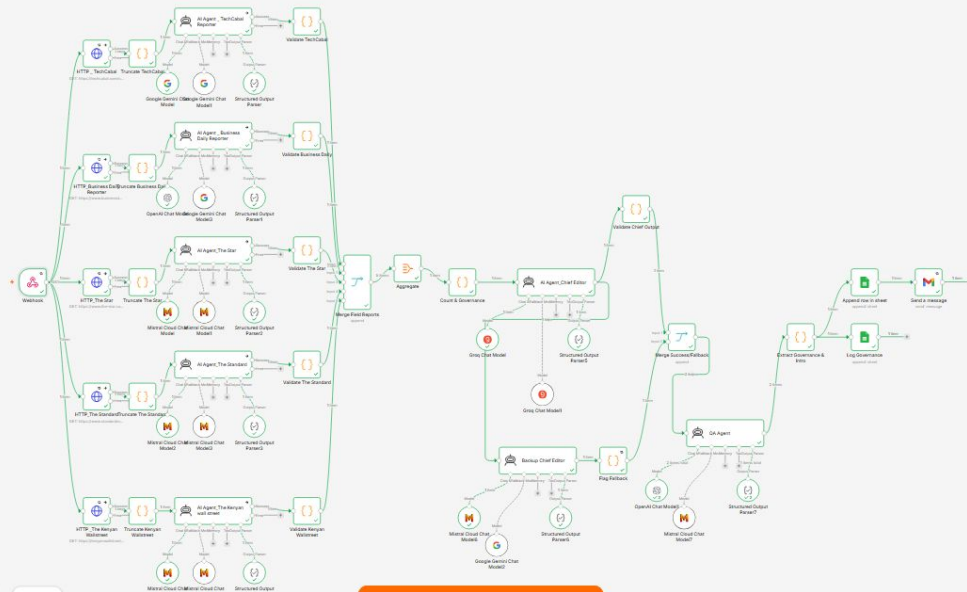
Publish



Editor

Executions

Evaluations



Execute workflow

Logs

Activate Windows

Go to Settings to activate Windows.

Clear execution



Search



00:32
08/06/2026



99+

Mail

Chat

Meet

Compose

Inbox595

Starred

Snoozed

Important

Sent

Drafts117

Purchases100

Social5,053

Updates28,138

Forums6

Promotions20,711

More

Labels

Search mail

Active

3 of 2,056

Nairobi Nexus Draft – 2026-06-08Inbox x

Amos Mokuia<amosmokuia3@gmail.com>

to me

00:28 (23 minutes ago)

Star

Emoji

Reply

More

Your Nairobi Nexus Weekly Digest

Niaje Bazeng! Sasa, let's dive into the latest tech and business news from Kenya. Kenya varsity students are winning big in global ICT competitions, and local fintech startup Pezesha has secured \$11 million in debt funding to scale its AI-powered lending platform. Form ni gani? Stay tuned for more updates on the Kenyan economy, business, and markets news, analysis, and data from Kenya and Africa.

This week's parting thought:

"May the power of innovation and entrepreneurship continue to shape a brighter future for Kenya and beyond."

Reply

Forward

Emoji

Activate Windows
Go to Settings to activate Windows.

Windows taskbar with icons for Start, Search, Task View, and various applications. System tray shows time 00:52 and date 08/06/2026.

Demo – Audit (Answer “What happened last Friday?”)

1. Manager opens the `audit.html` form, picks a date (e.g., 2026-06-01).
2. Clicks “Get Report”.
3. Within seconds, an email arrives: **“On 2026-06-01, 4 out of 5 sources succeeded. The Star failed. The model used was llama-3.3-70b-versatile. QA passed. The parting thought was ‘Keep learning, keep growing.’”**

****Result:**** Complete observability – anyone can ask about any execution.

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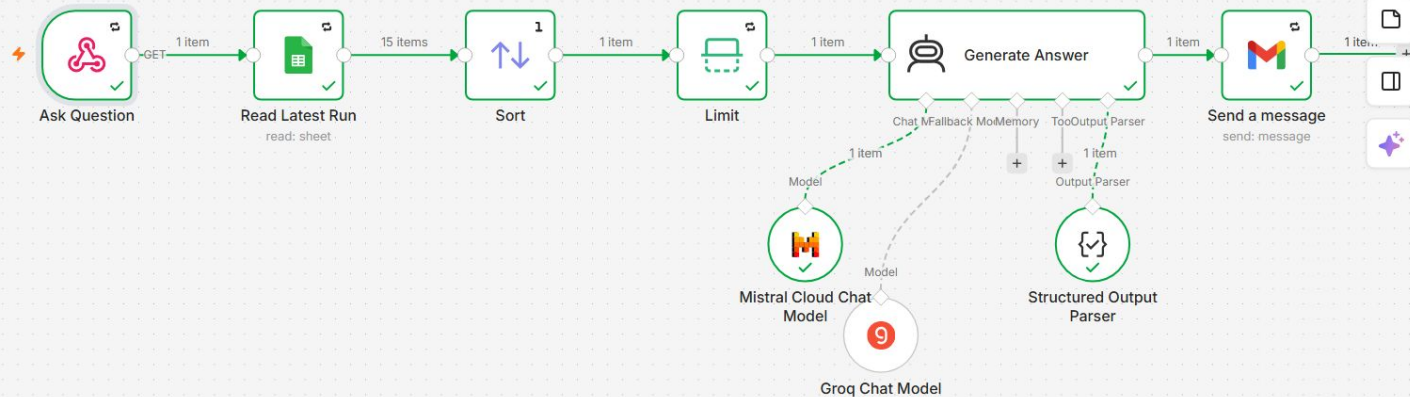
Personal / Audit Answer with AI + Add tag

Editor

Executions

Evaluations

Published



Execute workflow

Logs

Activate Windows

Go to Settings to activate Windows.

Clear execution



Search



00:37

08/06/2026

99+

Mail

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Important

Sent

Drafts117

Purchases100

Social5,053

Updates28,138

Forums6

Promotions20,711

More

Labels

Search mail

Active

2 of 2,056

Audit Answer: What happened last Friday?

Inbox x

Amos Mokua <amosmokua3@gmail.com>

to me

00:37 (15 minutes ago)

On June 4, 2026, the weekly newsletter was successfully generated using the llama-3.3-70b-versatile model, with all 5 sources reporting success and QA passing without issues.

This email was sent automatically with [n8n](#)

Reply

Forward

Activate Windows

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Demo – Fallback (AI Failure)

1. Temporarily break the primary model (invalid API key).
2. Workflow continues → orange error output triggers backup AI.
3. Email contains a shorter, simpler intro (apology, promise of full newsletter next week).
4. Governance sheet logs `Fallback Triggered = true` and the fallback model name.

****Result:**** System never fails completely – it degrades gracefully.

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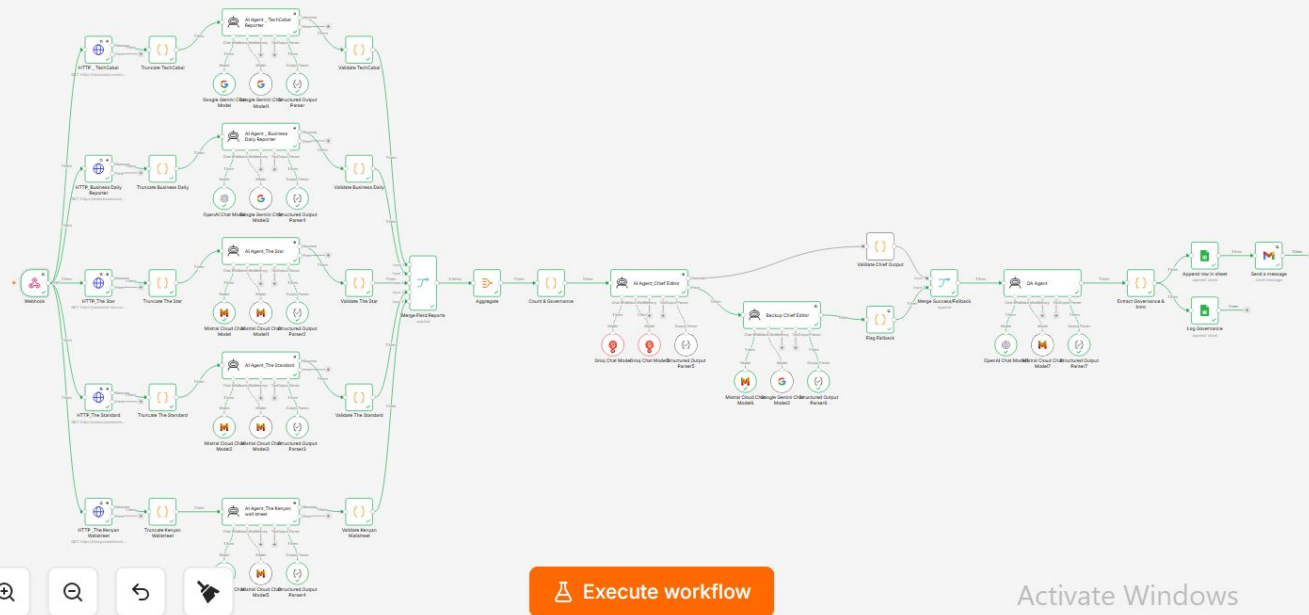
Personal / Main Workflow – Test Backup AI + Add tag

Published

Editor

Executions

Evaluations



Execute workflow

Activate Windows

Go to Settings to activate Windows.

Logs

Clear execution



Search

00:39
08/06/2026

99+

Mail

Chat

Meet

Compose

Inbox 595

Starred

Snoozed

Important

Sent

Drafts 117

Purchases 100

Social 5,053

Updates 28,138

Forums 6

Promotions 20,711

More

Search mail

Active

1 of 2,058

Nairobi Nexus Draft – 2026-06-08

Inbox x

Amos Mokua <amosmokua3@gmail.com>

00:59 (0 minutes ago)

Your Nairobi Nexus Weekly Digest

Sasa! Apologies for the short message, msee! The editor is MIA this week, but don't worry—we'll be back with a full, juicy newsletter next week. Stay tuned and keep it poa! 🤔 Catch you later!

💡 *This week's parting thought:*
"May the coming week bring you wisdom, peace, and prosperity."

This newsletter was automatically generated by the Nairobi Nexus AI system.

Reply

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Demo – Global Error Workflow (Fatal Failure)

1. Force a fatal error – e.g., insert a Code node with `throw new Error("Forced failure")` in a copy of the main workflow (or temporarily break an HTTP node and set On Error to Stop Workflow).
2. Trigger the workflow – call the production webhook.
3. Observe – The main workflow fails (red overall status).
4. Error workflow runs – The Error Trigger node fires, AI Agent generates a hypothesis and recommended action, and an email is sent to the ops team.
5. Email received – Subject:  Nairobi Nexus Newsletter – Workflow Failed at ..., Body contains hypothesis and action.

Result: Any fatal error is caught, analysed, and reported automatically – no silent failures.

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Personal Main workflow-Testing Error workflow + Add tag

Published

Executions No active executions

Auto refresh

Jun 7, 17:25:00

Error in 3.467s

Jun 7, 17:21:25

Error in 3.524s

Jun 7, 17:19:51

Error in 3.541s

Jun 7, 17:15:36

Error in 3.565s

Jun 7, 17:15:10

Error in 3.458s

Jun 7, 17:14:50

Succeeded in 45ms

Jun 7. 16:03:59

Which executions is this workflow saving?

Editor

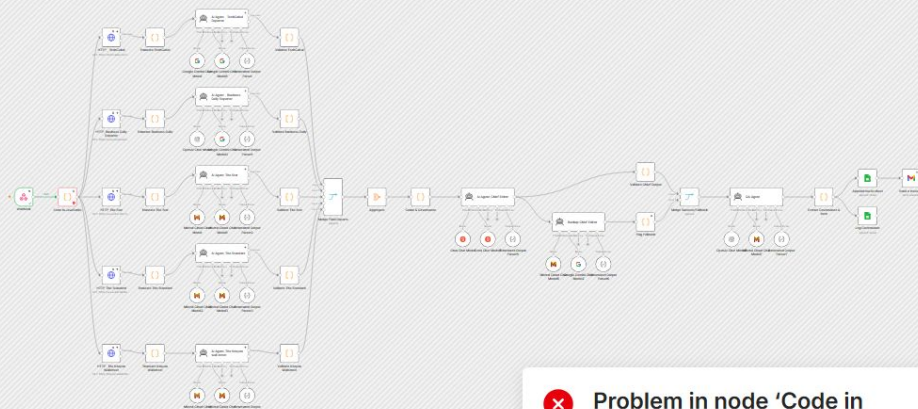
Executions

Evaluations

Jun 7, 17:25:00

Error in 3.467s | ID#897 | Version acb862ad

Debug in editor



Problem in node 'Code in JavaScript'

Forced test of global error workflow
[line 1]



Search



00:45
08/06/2026

n8n

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Personal / Error Workflow for NewsletterManager + Add tag

Published

...

Executions No active executions

Auto refresh

Jun 7, 17:25:04
Succeeded in 2.139s

Jun 7, 17:21:29
Succeeded in 1.764s

Jun 7, 17:19:54
Succeeded in 1.971s

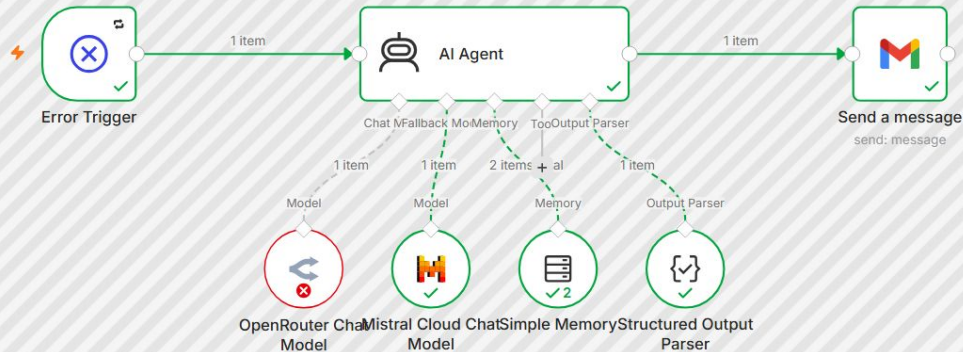
Jun 6, 20:22:54
Succeeded in 1.761s

Editor Executions Evaluations

Jun 7, 17:25:04

Succeeded in 2.139s | ID#898 | error workflow

Copy to editor



Activate Windows
Go to Settings to activate Windows.

Which executions is this workflow saving?

Logs



Search



00:45
08/06/2026

99+

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Search mail

Active

1 of 2,057

Nairobi Nexus Newsletter – Workflow Failed at

Inbox x

Amos Mokua <amosmokua3@gmail.com>

to me

00:55 (0 minutes ago)

Star

Emoji

Reply

More

Workflow Failure Alert

Hypothesis: The JavaScript code in the failing node might be explicitly triggering the global error workflow due to an unhandled error or intentional forced error test.

Recommended Action: Review the JavaScript code in the failing node to identify any forced error triggers or unhandled exceptions, and add proper error handling or logging to debug the issue.

Failed workflow: Main workflow-Testing Error workflow

Reply

Forward

Emoji

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Why It's Production-Grade

- ****No single point of failure**** – HTTP errors, AI failures, memory issues are all handled.
- ****Self-validating**** – defensive checks and QA ensure clean data always flows.
- ****Fully auditable**** – Governance sheet and audit workflow answer “What happened?”
- ****Human-centric**** – parting thought adds a personal touch, turning automation into connection.

Conclusion

- We transformed a prototype into a resilient, explainable, and thoughtful system.
- All Week 10 rubric criteria are met:
 -  Structured AI output & JSON reliability
 -  Fault tolerance & resilient design
 -  Model selection, constraints & fallbacks
 -  Operational logging & governance
- The system is ready for production and can be used by Nairobi Nexus every Friday.

****Thank you! Questions?****